- Broadcasting capabilities for handling operations on arrays of different shapes.
- A variety of mathematical functions for advanced computations.

2. Pandas

Pandas simplifies data manipulation and preprocessing by introducing data structures like Series and DataFrames, which make it easy to work with structured datasets. It supports reading and writing data from multiple file formats, including CSV, Excel, SQL, and JSON, making it versatile for real-world data handling. With Pandas, you can clean data, handle missing values, filter datasets, and perform grouping and aggregation, which are all essential steps in preparing data for ML models. Pandas' integration with NumPy and Matplotlib makes it an indispensable part of any Python data analysis workflow.

Example:

```
import pandas as pd
data = {"Name": ["Alice", "Bob", "Charlie"], "Score": [85, 90, 95], "Age": [24, 27, 22]}
df = pd.DataFrame(data)
print("Data Summary:\n", df.describe()) print("Filtered Data (Score > 88):\n", df[df['Score'] > 88])
df['Score'] += 5 # Adding bonus points print("Updated Scores:\n", df)
```

Features:

- Data wrangling with filtering, grouping, and merging datasets.
- Handling missing values and cleaning data.
- Supports time-series data manipulation for complex datasets.

3. Matplotlib

Matplotlib is a versatile library for creating static, animated, and interactive visualizations. It provides a high degree of control over plot elements, allowing developers to customize their visualizations extensively. From simple line plots to complex multi-layered graphs, Matplotlib serves as a powerful tool for exploratory data analysis. Its tight integration with Pandas and NumPy ensures seamless workflows, enabling quick visualization of data trends, patterns, and relationships.